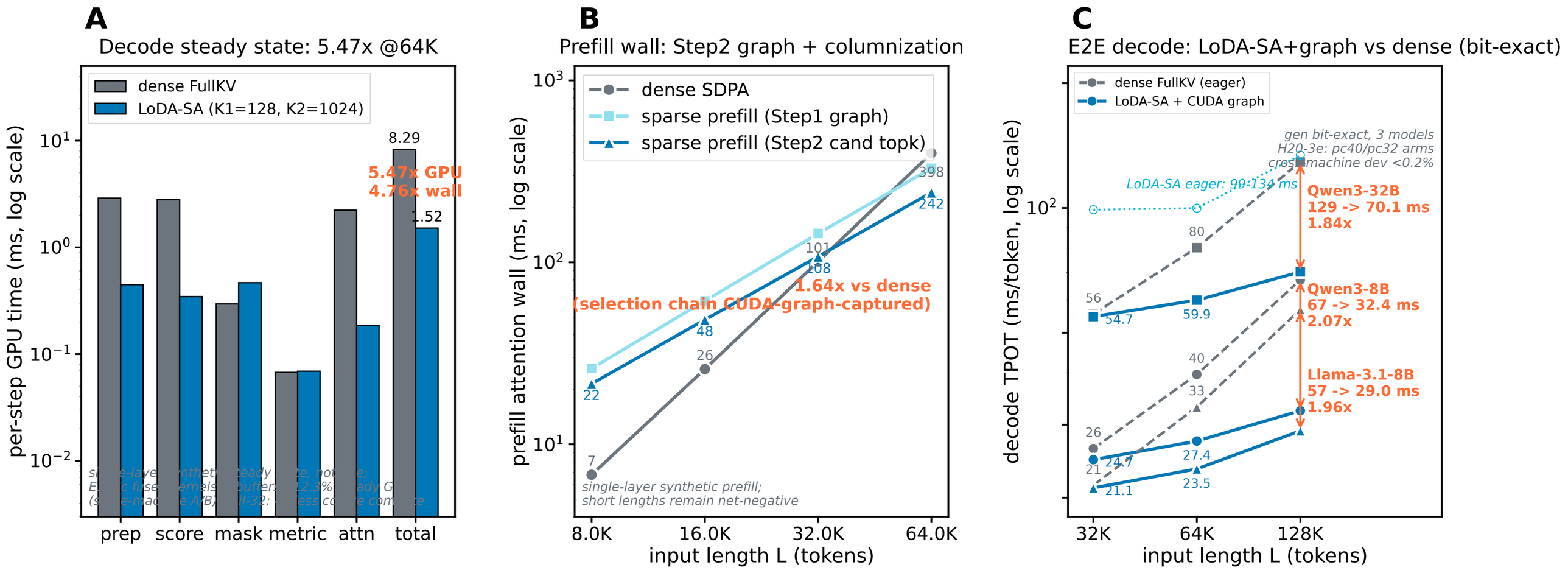
A
Decode steady state: 5.47x @64K
dense FullKV
LoDA-SA (K1=128, K2=1024)
per-step GPU time (ms, log scale)
8.29
1.52
5.47x GPU
4.76x wall
prep
score
mask
metric
attn
total
B
Prefill wall: Step2 graph + columnization
dense SDPA
sparse prefill (Step1 graph)
sparse prefill (Step2 cand topk)
prefill attention wall (ms, log scale)
1.64x vs dense
(selection chain CUDA-graph-captured)
single-layer synthetic prefill;
short lengths remain net-negative
8.0K
16.0K
32.0K
64.0K
input length L (tokens)
C
E2E decode: LoDA-SA+graph vs dense (bit-exact)
dense FullKV (eager)
LoDA-SA + CUDA graph
gen bit-exact, 3 models
H20-3e: pc40/pc32 arms
cross-machine dev <0.2%
LoDA-SA eager: 99-134 ms
Qwen3-32B
129 -> 70.1 ms
1.84x
Qwen3-8B
67 -> 32.4 ms
2.07x
Llama-3.1-8B
57 -> 29.0 ms
1.96x
decode TPOT (ms/token, log scale)
32K
64K
128K
input length L (tokens)